International Journal of Surgery Case Reports 127 (2025) 110988

Contents lists available at ScienceDirect

International Journal of Surgery Case Reports

journal homepage: www.elsevier.com/locate/ijscr

Case report

Case report of an omental metastasis of melanoma in a patient with abdominal pain

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ARTICLE INFO

Keywords:
Case report
Choroidal melanoma
Omental metastasis
Spindle cell melanoma
Metastatic melanoma
Surgical treatment

ABSTRACT

Introduction: Choroidal melanoma primarily metastasizes to the liver, but rare sites like the omentum can also be affected, making this case educational due to its uncommon presentation. The decision to pursue surgical treatment for metastatic melanoma remains controversial.

Case presentation: A woman in her 60s, with a history of a stable choroidal nevus, experienced rapid lesion growth, leading to enucleation and radiotherapy. Two years later, she developed severe epigastric pain. Imaging revealed peritoneal nodules, and histopathology confirmed metastatic choroidal melanoma. Molecular analysis identified a GNAQ G48L mutation.

Discussion: Omental metastasis in choroidal melanoma is rare, and while surgery is uncommon for metastatic melanoma, it was performed to alleviate symptoms and improve functionality. Following omentectomy and immunotherapy, her condition remained stable for two years.

Conclusion: This case emphasizes the importance of recognizing atypical metastatic sites in melanoma. Surgical intervention, though rare, can be beneficial for symptom relief in selected cases, improving outcomes when combined with targeted treatments.

1. Introduction

Choroidal melanoma is the most common primary intraocular malignancy in adults, representing 5 % of all human melanomas [1]. There are no clear epidemiological data on metastasis patterns in choroidal melanoma. However, an Argentinian cohort in 2024 studied 143 patients with choroidal melanoma. Metastases occurred in 19.6 % of cases, primarily targeting the liver in 46.4 % of cases, and less frequently in the lungs and bones [2]. Omental metastasis has been exceptionally described, making this case educational due to its unusual presentation [3]. Such cases challenge the typical understanding of metastatic patterns and broaden the scope of surveillance for clinicians managing melanoma patients. Recognizing and addressing these atypical metastatic sites is critical for timely intervention and improved patient outcomes.

Surgery can generally be done for isolated metastases that cause significant symptoms or functional impairment. Systemic therapies, such as immunotherapy or targeted therapies, remain the main treatment of metastatic melanoma management due to their survival benefits [4].

In this case report, following SCARE guidelines [5], we describe the decision-making process and outcomes of omentectomy performed to manage a rare metastatic presentation of choroidal melanoma.

This patient was treated at the University Hospital in France, an academic medical center with a multidisciplinary team of oncologists, surgeons, and radiologists. The institution's resources allowed for precise imaging, molecular diagnostics, and collaborative decision-making.

2. Case presentation

2.1. Initial diagnosis period

A 64-year-old woman presented with a known history of choroidal nevus in the left eye, which had been stable for several years. She had no known comorbidities or family history of melanoma or other cancers. In June 2019, a significant increase in the size of the nevus was observed over a three-month interval. Orbital MRI revealed scleral extension of the lesion (Fig. 1).

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https://doi.org/10.1016/j.ijscr.2025.110988
Received 18 November 2024; Received in revised form 27 January 2025; Accepted 28 January 2025

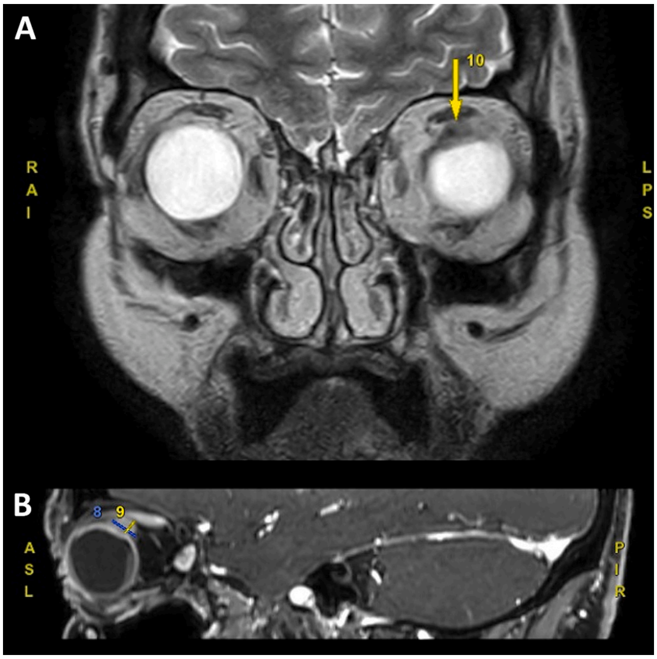

Fig. 1. Frontal view (A) and parasagittal view (B) of an orbital MRI showing scleral extension of the choroidal melanoma.

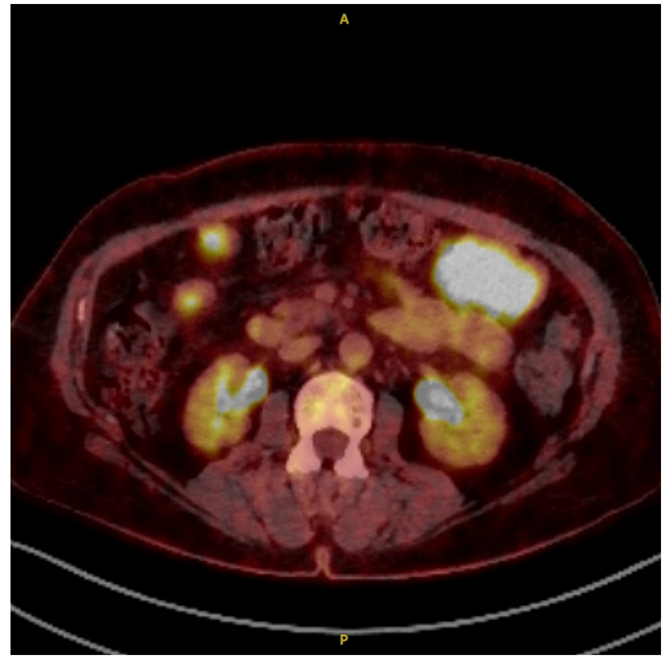

Fig. 2. PET-CT scan of the abdomen and pelvis revealing peritoneal nodules suggestive of metastatic melanoma.

Initially, the patient underwent enucleation of the left eye. Excision was complete. Histopathological analysis confirmed a choroidal melanoma of the spindle cell type, measuring 2 × 0.7 cm, with scleral protrusion (Fig. 4A). BAP1 (BRCA1-associated protein-1) expression was preserved. The tumor was staged as pT4e Nx (TNM 8th edition), with no lymphovascular invasion (L0 V0 R0). The patient underwent afterwards stereotactic radiotherapy to the left orbit.

2.2. Metastatic period

The patient was treated with trametinib at a dose of 1.5 mg per day, resulting in and disease control over two years. Two years after initial surgery, the patient reported severe epigastric pain. An initial computed tomography (CT) scan of the abdomen revealed multiple peritoneal nodules. To further evaluate the extent and activity of the lesions, a positron emission tomography-computed tomography (PET-CT) scan was performed and revealed peritoneal nodules (max 27 mm) suggestive of metastatic choroidal melanoma (Fig. 2). Molecular analysis identified a GNAQ G48L mutation, with negative results for B-RAF, N-RAS, C-KIT, and NTRK mutations.

The patient underwent omentectomy, and histopathology confirmed the presence of omental metastasis from the choroidal melanoma (Figs. 3 and 4B, C, D). The excised omentum displayed multiple dark pigmented lesions consistent with melanoma metastases (Fig. 3). Following surgery and immunotherapy, the patient's abdominal pain regressed, and her condition has remained stable for the past two years.

3. Discussion

Choroidal melanoma is the most common primary intraocular malignancy in adults and has a high propensity for metastasis, primarily to the liver. Several factors may explain why metastasis to the omentum might occur such as hematogenous and lymphatic spread, tumor characteristics, and microenvironment suitability [6]. The omentum is known to act as a metastatic niche for some cancers due to its rich vascularization, high lymphatic drainage and abundant adipose tissue providing ideal microenvironment for survival and growth of metastatic cells [7].

Fig. 3. Excised omentum displaying multiple dark pigmented lesions consistent with melanoma metastases.

Histopathological analysis of the primary tumor revealed a spindle cell choroidal melanoma with preserved BAP1 in immunochemistry. This is noteworthy as loss of BAP1 expression is often associated with a higher risk of metastasis and worse overall survival outcomes in uveal melanoma [8]. Molecular analysis of the omentum identified a GNAQ mutation, which is found in 80 to 90 % of uveal melanomas. This mutation leads to constitutive activation of the MAPK and PI3K signaling pathways, promoting tumorigenesis and progression [9].

The images clearly show the typical pigmentation of melanoma, helping for differential diagnosis.

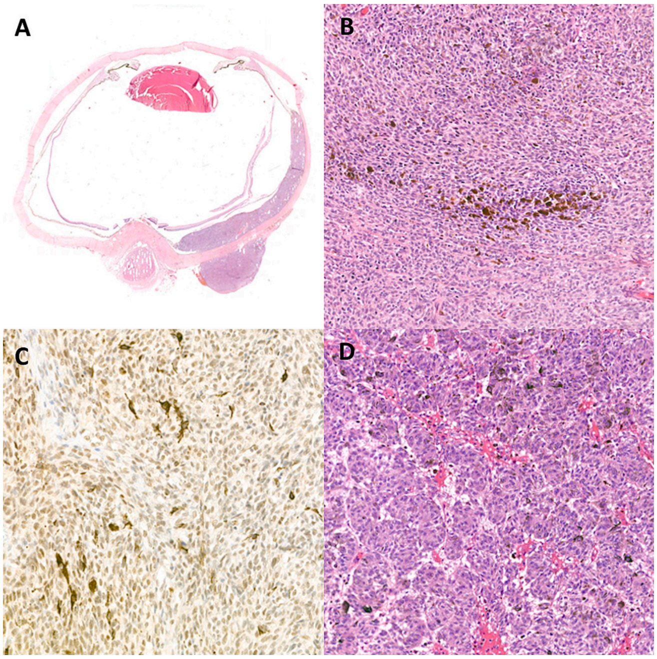

Fig. 4. Histopathologic analysis of choroidal melanoma with scleral invasion (A). Hematoxylin and eosin stain revealed tightly packed atypical spindle cells mixed with melanin pigments (B). Selective staining for BAP1 was retained (C). Histopathologic analysis of the omental metastasis, H&E stain showing nests of atypical spindle cells with melanin pigments (D).

In this case, the decision to proceed with omentectomy was driven by the need for symptom relief and improvement in quality of life, rather than curative intent. This highlights the role that surgery can play in selected metastatic cases, where functional outcomes are prioritized. Due to the rarity of this metastatic localization, there are no established treatment guidelines, highlighting the critical role of tumor board meetings and a multidisciplinary approach. In this case, surgery was performed to address the functional impact, demonstrating its importance in managing symptoms. Metastatic melanoma may benefit from metastasectomies, as suggested by a retrospective study of 226 patients treated between 1999 and 2020, when combined with targeted treatments [10], or as shown in a matched-pair analysis of 2353 patients where surgery combined with systemic therapies improved 5-year specific survival compared to systemic therapy alone (58.8 % vs. 38.9 %, $p < 0.05$) [11]. However, further studies are needed to determine whether surgical intervention influences survival in our case which would aid in making more informed benefit-risk assessments for treating this exceptional metastatic site for uveal melanomas.

4. Conclusion

This case highlights the importance of recognizing atypical metastatic sites, such as the omentum, in patients with choroidal melanoma. While metastasis to the omentum is exceptionally rare, it underscores the need for vigilance and comprehensive imaging in the surveillance of melanoma patients. The case also shows the potential role of surgery in managing symptoms, potentially improving quality of life. Multidisciplinary collaboration involving oncologists, surgeons, radiologists, and pathologists is critical in optimizing patient care and ensuring a tailored treatment approach. Further studies are needed to assess the long-term impact of surgical intervention on survival in rare metastatic sites like the omentum.

CRediT authorship contribution statement

Jonathan Sabah = study concept, original draft writing, photography collection
Alexis Marouk = case writing and reviewing
Louis Vallois = photography collection, case writing and reviewing
Chérif Akladios = study concept.

All authors read and approved the manuscript.

Ethical approval

This research was conducted ethically in accordance with the Declaration of Helsinki. The patient has given her written informed and oral consent to publish the case (including image publication).

Guarantor

Jonathan Sabah, MD.

Research registration number

This study did not need registration of Research Studies since it is not a First in Man case report.

Funding

There is no funding for this article.

Declaration of competing interest

The authors state that they have no conflict of interest.

References

[1] A.D. Singh, M.E. Turell, A.K. Topham, Uveal melanoma: trends in incidence, treatment, and survival, Ophthalmology 118 (9) (2011) 1881–1885, https://doi.org/10.1016/j.ophtha.2011.01.040.
[2] C.M. Gentile, Y. Müller, L. Tajtelbaum, A. Lombardi, J.O. Croxatto, Incidencia de metástasis y sobrevida en pacientes con melanoma uveal primario [Incidence of metastasis and survival in patients with primary uveal melanoma], Medicina (B Aires) 84 (5) (2024) 902–912 (Spanish, PMID: 39399930).
[3] H. Zhao, M. Yang, X. Zhang, S. Zhao, G. Yao, D. Zhou, et al., The first case of omental metastasis from primary choroidal melanoma, Jpn. J. Clin. Oncol. 43 (3) (2013) 314–317, https://doi.org/10.1093/jjco/hys229 (PMID: 23293369).
[4] I. Mirzayev, A.K. Gündüz, L. Mirzayeva, K. Ceyhan, Extended survival in a case of metastatic choroidal melanoma with immunotherapy, Anti-Cancer Drugs (Jul 30 2024), https://doi.org/10.1097/CAD.0000000000001650 (Epub ahead of print. PMID: 39079167).
[5] C. Sohrabi, G. Mathew, N. Maria, A. Kerwan, T. Franchi, R.A. Agha, The SCARE 2023 guideline: updating consensus Surgical CAse REport (SCARE) guidelines, Int J Surg Lond Engl. 109 (5) (2023) 1136.
[6] L. Calomarde-Rees, R. García-Calatayud, C. Requena Caballero, E. Manrique-Silva, V. Traves, Z. García-Casado, et al., Risk factors for lymphatic and hematogenous dissemination in patients with stages I to II cutaneous melanoma, JAMA Dermatol. 155 (6) (2019) 679–687, https://doi.org/10.1001/jamadermatol.2019.0069 (PMID: 31042264; PMCID: PMC6495361).
[7] P. Iyengar, T.P. Combs, A.J. Dannenberg, X.K. Zhou, A. Gucalp, K. Subbaramaiah, et al., Omentum and bone marrow: how adipocyte-rich organs create tumour microenvironments conducive for metastatic progression, Obes. Rev. 17 (11) (2016) 1114–1128, https://doi.org/10.1111/obr.12450.
[8] S. Kennedy, S. Owens, L. Ivers, C. Hegarty, V. O'Neill, J.J. Berenguer-Pina, et al., Prognostic value of BAP1 protein expression in uveal melanoma, Am. J. Surg. Pathol. 48 (3) (2024) 329–336, https://doi.org/10.1097/PAS.0000000000002176 (PMID: 38238977; PMCID: PMC10876168).
[9] A.R. Moore, E. Ceraudo, J.J. Sher, Y. Guan, A.N. Shoushtari, M.T. Chang, et al., Recurrent activating mutations of G-protein-coupled receptor CYSLTR2 in uveal melanoma, Cancers (Basel) 11 (6) (2019) 846, https://doi.org/10.3390/cancers11060846.
[10] I.R. Schwantes, T. Sutton, S. Behrens, G. Fowler, G. Han, J.T. Vetto, et al., Metastasectomy for metastatic melanoma in the era of effective systemic therapy, Am. J. Surg. 226 (6) (2023) 1234–1242, https://doi.org/10.1016/j.amjsurg.2023.04.020.
[11] D.W. Nelson, T.D. Fischer, A.N. Graff-Baker, A. Dehal, S. Stern, A.J. Bilchik, et al., Impact of effective systemic therapy on metastasectomy in stage IV melanoma: a

matched-pair analysis, Ann. Surg. Oncol. 26 (2019) 4610–4618, https://doi.org/10.1245/s10434-019-07487-5.